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Master of Computer Applications

Semester – III

Data Analytics for Business

Final Project

Title: RETAIL BUSINESS ANALYTICS AND SALES PERFORMANCE DASHBOARD

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RETAIL BUSINESS ANALYTICS AND SALES PERFORMANCE DASHBOARD

1. ABSTRACT

The **Retail Business Analytics and Sales Performance Dashboard** is a data analytics project developed to help retail organizations understand sales performance, customer behavior, product performance, and geographical trends.

The project integrates retail transaction data with **MySQL** for structured storage and data processing and **Microsoft Power BI** for interactive visualization and business intelligence. The raw sales dataset is imported into MySQL, cleaned, transformed, and organized into a star-schema data warehouse consisting of fact and dimension tables.

Power BI is then connected to the MySQL database to create an analytical data model. DAX measures are developed to calculate important Key Performance Indicators (KPIs), including total sales, total orders, total customers, total quantity, average order value, sales growth, and year-to-date sales.

The final solution provides interactive dashboards for executives, sales teams, product managers, and business decision-makers. Additional features such as forecasting, What-If analysis, Row-Level Security, Power BI Service deployment, and GitHub-based project management make the project an end-to-end business analytics solution.

2. INTRODUCTION

Retail organizations generate large amounts of transactional data every day. This data contains valuable information about products, customers, orders, quantities, prices, countries, and sales dates.

However, raw transactional data is difficult for management to analyze directly. Businesses require a centralized analytics platform that can transform raw data into meaningful information.

This project develops a complete retail analytics solution using:

- MySQL Workbench
- SQL
- Power Query
- Power BI Desktop
- DAX
- Power BI Service
- GitHub

The system follows the complete data analytics lifecycle:

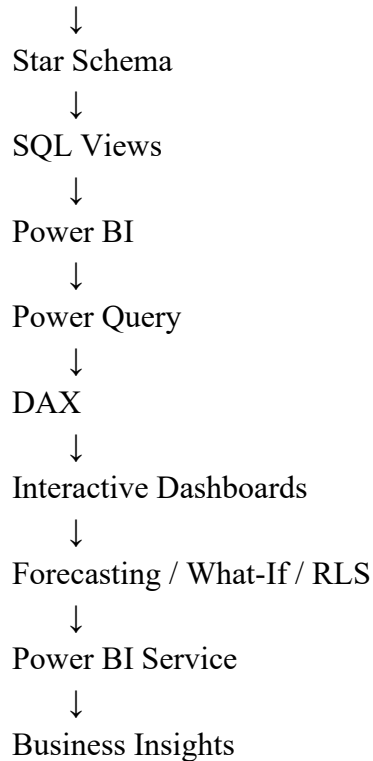
Raw CSV Dataset



MySQL Database



Data Cleaning



3. PROBLEM STATEMENT

Retail sales data is often stored in large CSV or Excel files containing hundreds of thousands of transaction records.

Analyzing such data manually creates several challenges:

- Difficult identification of sales trends
- Difficult comparison of products
- Limited customer analysis
- Difficulty analyzing geographical performance
- Manual calculation of KPIs
- Lack of interactive reporting
- Difficulty forecasting future sales
- Limited security for different business users

Therefore, there is a need for an integrated business analytics solution that can store, clean, model, analyze, visualize, and securely distribute retail sales information.

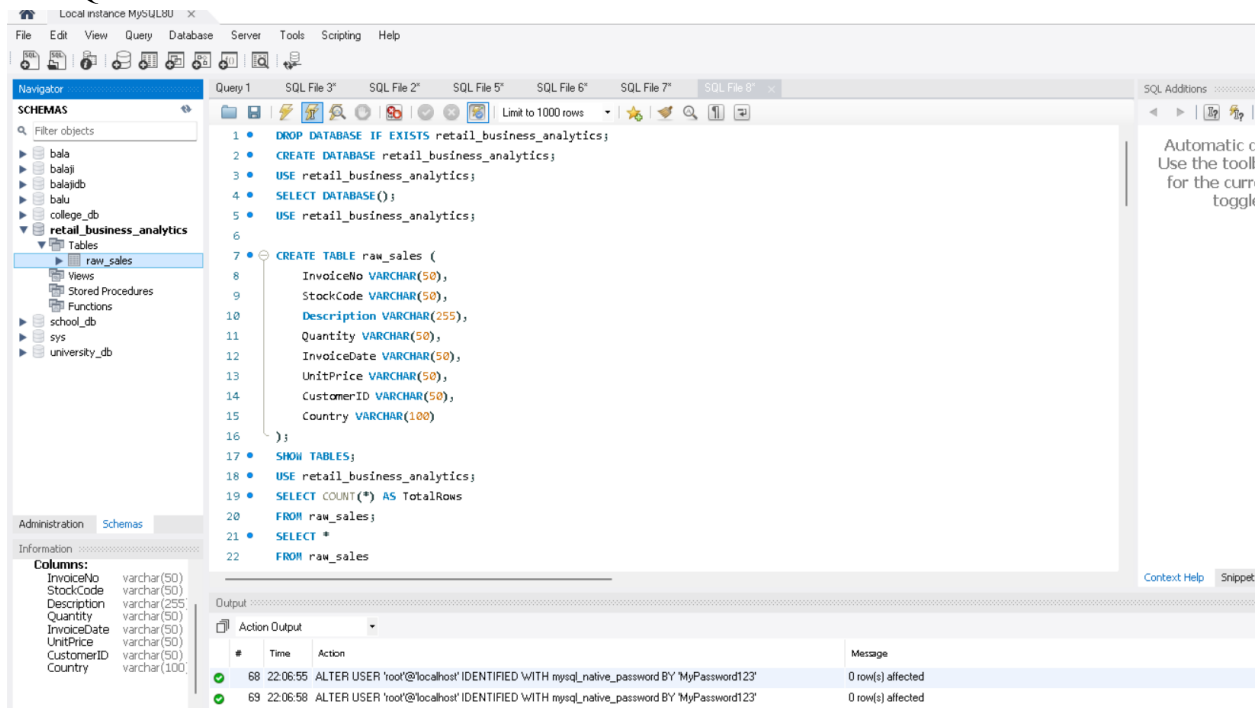
4. OBJECTIVES

The main objectives are:

- Import retail transaction data into MySQL.
- Clean and validate the raw dataset.
- Remove invalid transactions.
- Create a structured star-schema database.

- Create fact and dimension tables.
- Connect MySQL with Power BI.
- Develop DAX measures for business KPIs.
- Create interactive dashboards.
- Analyze sales performance by time and country.
- Identify top-performing products.
- Analyze customer performance.
- Implement sales forecasting.
- Implement What-If analysis.
- Implement Row-Level Security.
- Publish the report to Power BI Service.
- Maintain project files using GitHub.
- Generate actionable business insights.

MYSQL TO POWER BI SERVER CONNECTIONS



5. SCOPE OF THE PROJECT

The project covers:

Data Management

- CSV data import
- MySQL database
- Data cleaning
- Data validation

Data Modeling

- Fact table
- Product dimension
- Customer dimension
- Country dimension
- Date dimension

Analytics

- Sales analysis
- Product analysis
- Customer analysis
- Geographic analysis
- Trend analysis
- Forecasting

Business Intelligence

- KPI cards
- Charts
- Slicers
- Interactive reports
- What-If analysis
- RLS

Deployment

- Power BI Service
- Data refresh
- GitHub repository

6. EXISTING SYSTEM

In a traditional retail environment, sales information may be maintained in spreadsheets or separate files.

The existing approach can have:

- Manual reporting
- Multiple data sources
- Duplicate records
- Data quality problems
- Slow analysis
- Static reports
- Limited visualization
- Difficult KPI tracking

Management may need to manually combine different files before making business decisions.

7. PROPOSED SYSTEM

The proposed system creates a centralized analytics platform.

CSV



MySQL



Cleaning



Star Schema



Power BI



DAX



Interactive Dashboard



Power BI Service

The proposed solution allows users to:

- Monitor sales
- Track orders
- Analyze customers
- Compare countries
- Identify top products
- Analyze trends
- Forecast sales
- Perform scenario analysis
- Secure reports by user role

8. DATASET DESCRIPTION

The project uses a retail transaction dataset containing fields such as:

Column	Description
InvoiceNo	Transaction/invoice number
StockCode	Product code
Description	Product description
Quantity	Number of products sold
InvoiceDate	Date and time of transaction
UnitPrice	Price per unit

Column	Description
CustomerID	Customer identifier
Country	Customer country

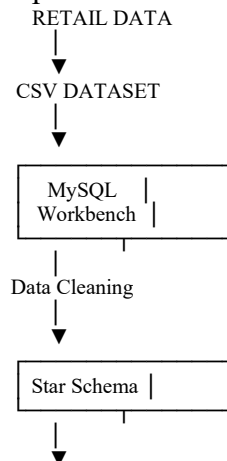
The dataset contains approximately 541,909 raw transaction records before cleaning, depending on the supplied CSV version.

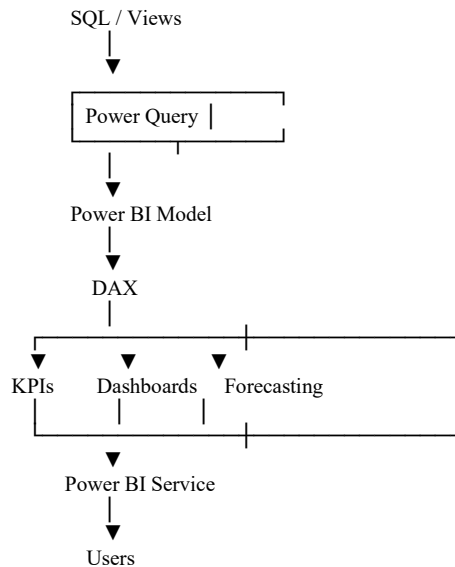
9. TECHNOLOGY STACK

Technology	Purpose
CSV	Source dataset
MySQL Workbench	Database management
SQL	Data cleaning and transformation
Power Query	Data preparation
Power BI Desktop	Dashboard development
DAX	Business calculations
Power BI Service	Report publishing
GitHub	Version control
Microsoft PowerPoint	Project presentation
PDF	Final documentation

10. SYSTEM ARCHITECTURE

The complete architecture is:





11. MYSQL DATABASE IMPLEMENTATION

11.1 Database Creation

```

DROP DATABASE IF EXISTS retail_business_analytics;
CREATE DATABASE retail_business_analytics;
USE retail_business_analytics;

```

11.2 Raw Sales Table

```

CREATE TABLE raw_sales (
    InvoiceNo VARCHAR(20),
    StockCode VARCHAR(30),
    Description VARCHAR(255),
    Quantity INT,
    InvoiceDate VARCHAR(30),
    UnitPrice DECIMAL(12,4),
    CustomerID VARCHAR(30),
    Country VARCHAR(100)
);

```

InvoiceDate is initially stored as VARCHAR to avoid date-import errors during the raw data-loading stage.

11.3 Data Import

The CSV file is imported using:

MySQL Workbench → Table Data Import Wizard

The following fields are mapped:

InvoiceNo
 StockCode
 Description
 Quantity
 InvoiceDate

UnitPrice
CustomerID
Country

11.4 Data Validation

The number of records is checked using:

```
SELECT COUNT(*) AS TotalRows
```

```
FROM raw_sales;
```

Sample records can be inspected using:

```
SELECT *
```

```
FROM raw_sales
```

```
LIMIT 10;
```

12. DATA CLEANING AND TRANSFORMATION

A clean dataset is necessary for reliable analytics.

12.1 Date Conversion

```
CREATE TABLE clean_sales AS
```

```
SELECT
```

```
    TRIM(InvoiceNo) AS InvoiceNo,
```

```
    TRIM(StockCode) AS StockCode,
```

```
    TRIM(Description) AS Description,
```

```
    Quantity,
```

```
    STR_TO_DATE(
```

```
        TRIM(InvoiceDate),
```

```
        '%m/%d/%Y %H:%i'
```

```
    ) AS InvoiceDate,
```

```
    UnitPrice,
```

```
    NULLIF(TRIM(CustomerID), '') AS CustomerID,
```

```
    TRIM(Country) AS Country
```

```
FROM raw_sales;
```

12.2 Invalid Transaction Removal

```
CREATE TABLE sales_final AS
SELECT *
FROM clean_sales
WHERE Quantity > 0
AND UnitPrice > 0
AND InvoiceDate IS NOT NULL;
```

This removes records with:

- Invalid quantities
- Invalid prices
- Invalid dates

12.3 Sales Amount Calculation

```
CREATE TABLE sales_data AS
SELECT
    InvoiceNo,
    StockCode,
    Description,
    Quantity,
    InvoiceDate,
    UnitPrice,
    CustomerID,
    Country,
    Quantity * UnitPrice AS SalesAmount
FROM sales_final;
```

The SalesAmount field represents:

Sales Amount = Quantity × Unit Price

12.3 Sales Amount Calculation

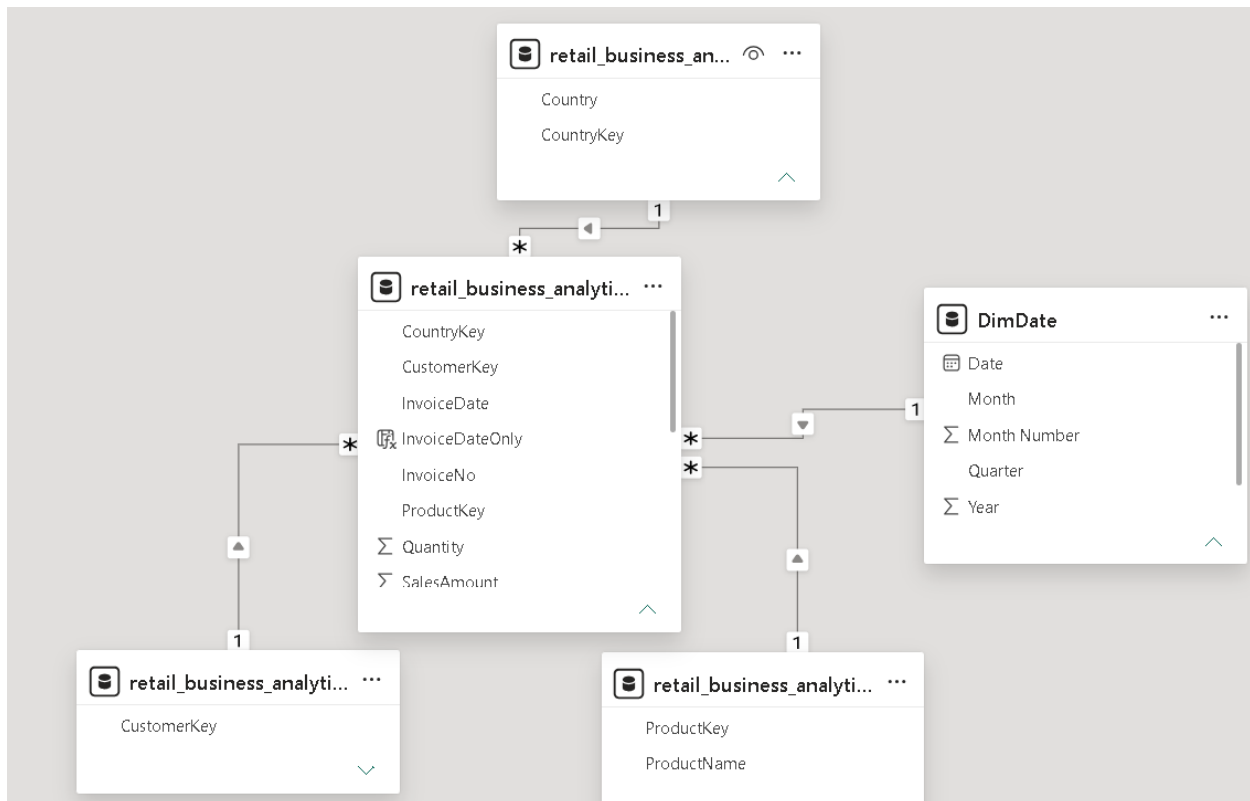
```
CREATE TABLE sales_data AS
SELECT
    InvoiceNo,
    StockCode,
    Description,
```

Quantity,
 InvoiceDate,
 UnitPrice,
 CustomerID,
 Country,
 Quantity * UnitPrice AS SalesAmount

FROM sales_final;

The SalesAmount field represents:

Sales Amount = Quantity × Unit Price



14. DATABASE TABLES

14.1 DimProduct

CREATE TABLE DimProduct (

ProductKey INT AUTO_INCREMENT PRIMARY KEY,

StockCode VARCHAR(30) NOT NULL,

```
    ProductName VARCHAR(255)  
);
```

14.2 DimCustomer

```
CREATE TABLE DimCustomer (  
    CustomerKey INT AUTO_INCREMENT PRIMARY KEY,  
    CustomerID VARCHAR(30) NOT NULL  
);
```

14.3 DimCountry

```
CREATE TABLE DimCountry (  
    CountryKey INT AUTO_INCREMENT PRIMARY KEY,  
    Country VARCHAR(100) NOT NULL UNIQUE  
);
```

14.4 FactSales

```
CREATE TABLE FactSales (  
    SalesKey BIGINT AUTO_INCREMENT PRIMARY KEY,  
    InvoiceNo VARCHAR(20),  
    InvoiceDate DATETIME,  
    ProductKey INT,  
    CustomerKey INT NULL,  
    CountryKey INT,  
    Quantity INT,  
    UnitPrice DECIMAL(12,4),  
    SalesAmount DECIMAL(14,4)  
);
```

15. POWER BI DATA MODEL

Power BI is connected to MySQL using:

Home → Get Data → MySQL Database

Database:

retail_business_analytics

Tables imported:

FactSales

DimProduct

DimCustomer

DimCountry

A separate DimDate table is created in Power BI.

Query [4] = Table.TransformColumns(#\"Cleaned Text1\",{{\"CountryKey\", Text.Trim, type text}})

InvoiceNo	ProductKey	CustomerKey	CountryKey	InvoiceDate	Quantity	UnitPrice
1 536365	85123A	17850	United Kingdom	01-12-2010 08:26:00	6	
2 536365	71053	17850	United Kingdom	01-12-2010 08:26:00	6	
3 536365	844068	17850	United Kingdom	01-12-2010 08:26:00	6	
4 536365	84029G	17850	United Kingdom	01-12-2010 08:26:00	6	
5 536365	84029E	17850	United Kingdom	01-12-2010 08:26:00	6	
6 536365	22752	17850	United Kingdom	01-12-2010 08:26:00	2	
7 536365	21730	17850	United Kingdom	01-12-2010 08:26:00	6	
8 536366	22633	17850	United Kingdom	01-12-2010 08:28:00	6	
9 536366	22632	17850	United Kingdom	01-12-2010 08:28:00	6	
10 536367	84879	13047	United Kingdom	01-12-2010 08:34:00	32	
11 536367	22745	13047	United Kingdom	01-12-2010 08:34:00	6	
12 536367	22748	13047	United Kingdom	01-12-2010 08:34:00	6	
13 536367	22749	13047	United Kingdom	01-12-2010 08:34:00	6	
14 536367	22310	13047	United Kingdom	01-12-2010 08:34:00	6	
15 536367	84969	13047	United Kingdom	01-12-2010 08:34:00	6	
16 536367	22623	13047	United Kingdom	01-12-2010 08:34:00	3	
17 536367	22622	13047	United Kingdom	01-12-2010 08:34:00	2	
18 536367	21754	13047	United Kingdom	01-12-2010 08:34:00	3	
19 536367	21755	13047	United Kingdom	01-12-2010 08:34:00	3	
20 536367	21777	13047	United Kingdom	01-12-2010 08:34:00	4	
21 536367	48187	13047	United Kingdom	01-12-2010 08:34:00	4	
22 536368	22960	13047	United Kingdom	01-12-2010 08:34:00	6	
23 536368	22913	13047	United Kingdom	01-12-2010 08:34:00	3	
24 536368	22912	13047	United Kingdom	01-12-2010 08:34:00	3	
25 536368	22911	13047	United Kingdom	01-12-2010 08:34:00	3	

16. DATE TABLE

DimDate =

ADDCOLUMNS(

 CALENDAR(

 DATE(2010,12,1),

 DATE(2011,12,31)

),

 "Year", YEAR([Date]),

 "Month Number", MONTH([Date]),

 "Month", FORMAT([Date], "MMMM"),

 "Year Month", FORMAT([Date], "YYYY-MM"),

 "Quarter", "Q" & FORMAT([Date], "Q")

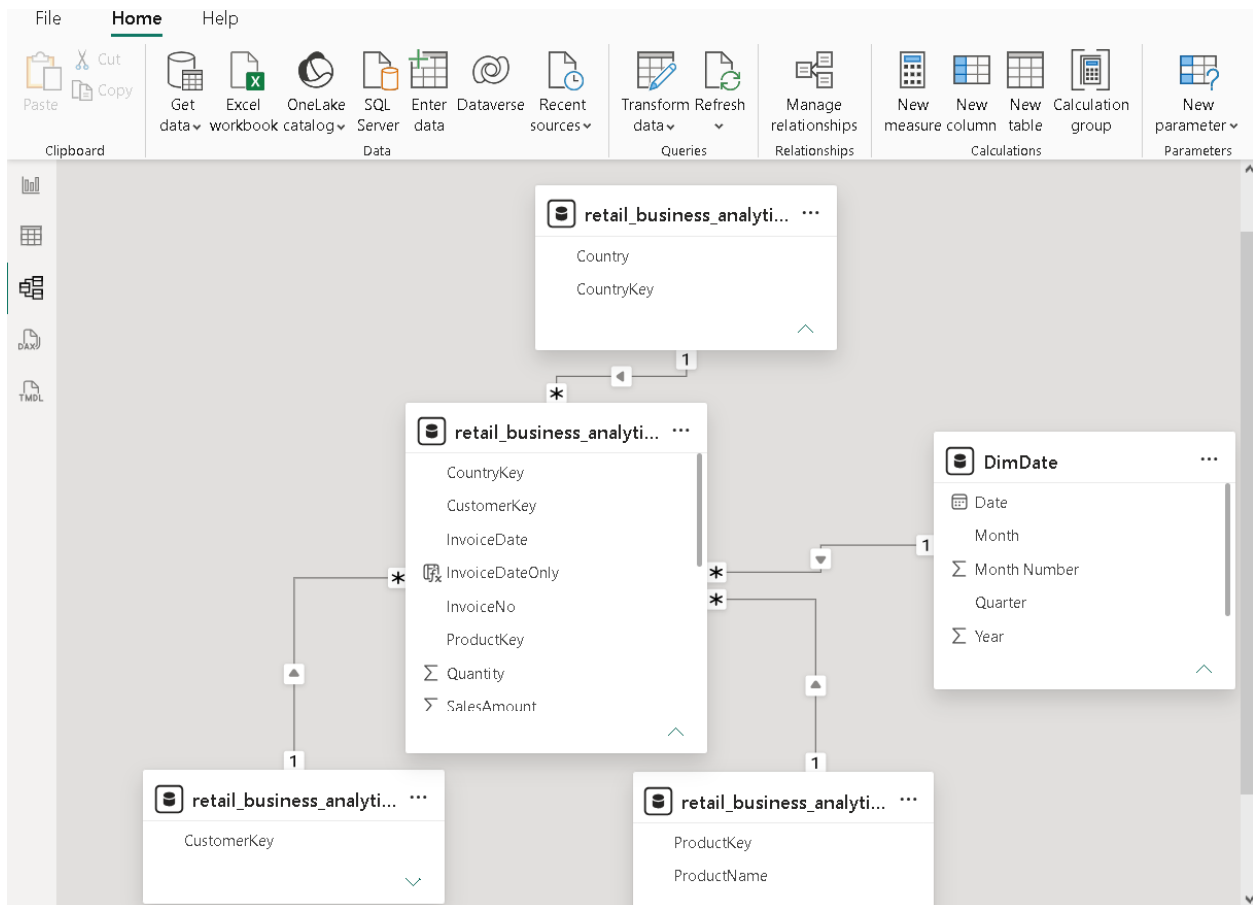
)

The date table enables:

- Year analysis
- Month analysis
- Quarter analysis
- YTD calculations
- Time-based comparisons

17. POWER BI RELATIONSHIPS

The model follows a one-to-many relationship structure.



18. DAX MEASURES

Total Sales

Total Sales =

SUM(FactSales[SalesAmount])

Total Orders

Total Orders =

DISTINCTCOUNT(FactSales[InvoiceNo])

Total Customers

Total Customers =

DISTINCTCOUNT(FactSales[CustomerKey])

Total Quantity

Total Quantity =

SUM(FactSales[Quantity])

Average Order Value

Average Order Value =

DIVIDE(

[Total Sales],

[Total Orders],

0

)

Average Unit Price

Average Unit Price =

AVERAGE(FactSales[UnitPrice])

Sales per Customer

Sales per Customer =

DIVIDE(

[Total Sales],

[Total Customers],

0

)

Previous Year Sales

Previous Year Sales =

CALCULATE(

[Total Sales],

SAMEPERIODLASTYEAR(DimDate[Date])

)

Sales Growth

Sales Growth % =

DIVIDE(

[Total Sales] - [Previous Year Sales],

[Previous Year Sales],

0

)

YTD Sales

YTD Sales =

TOTALYTD(

[Total Sales],

DimDate[Date]

)

19. DASHBOARD DESIGN

The dashboard uses a consistent professional theme.

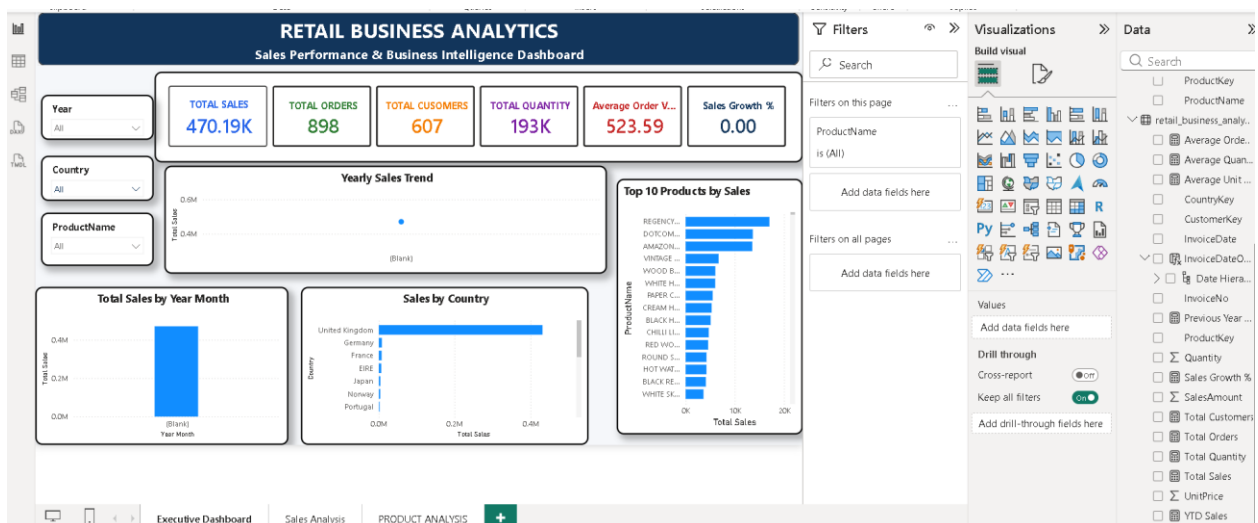
20. PAGE 1 — EXECUTIVE DASHBOARD

Title

RETAIL BUSINESS ANALYTICS

Subtitle

Executive Sales & Business Performance



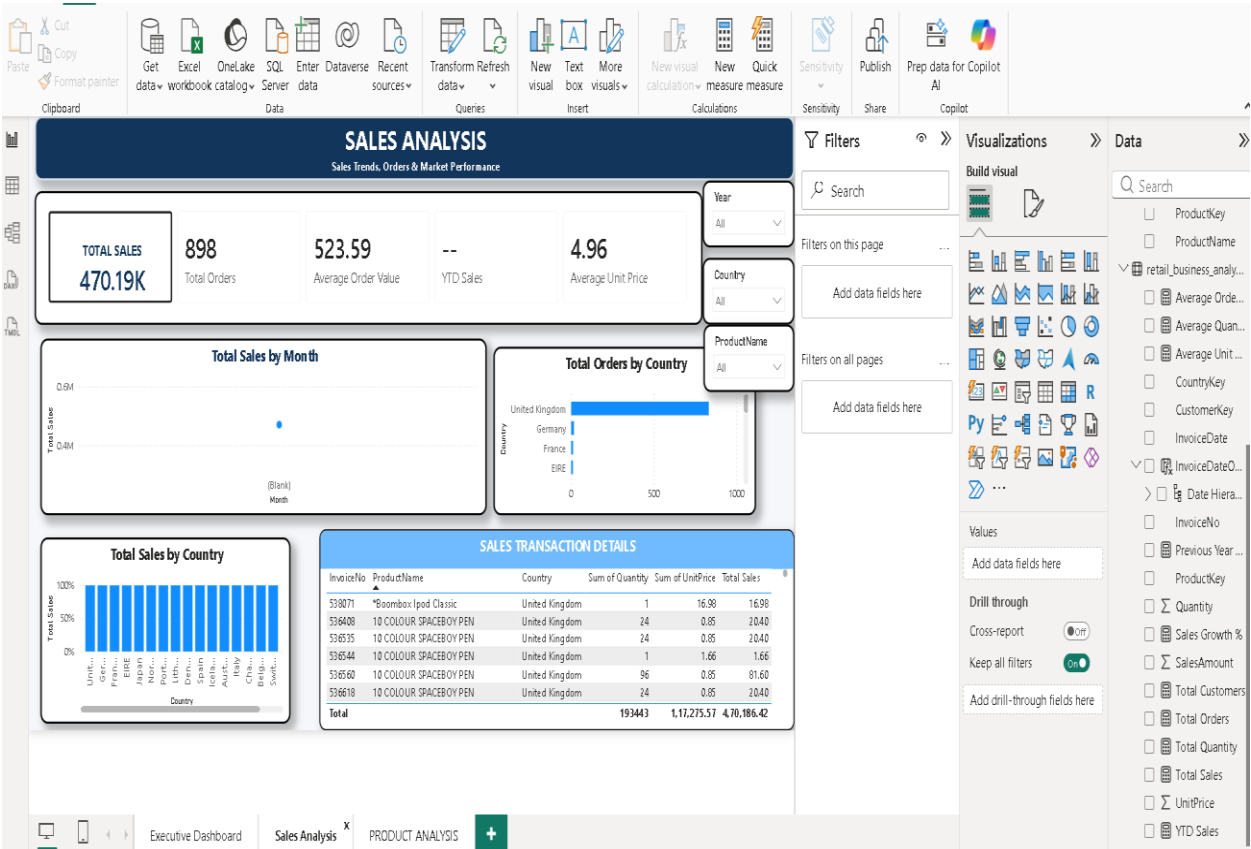
21. PAGE 2 — SALES ANALYSIS

Title

SALES ANALYSIS

Subtitle

Sales Trends, Orders & Market Performance



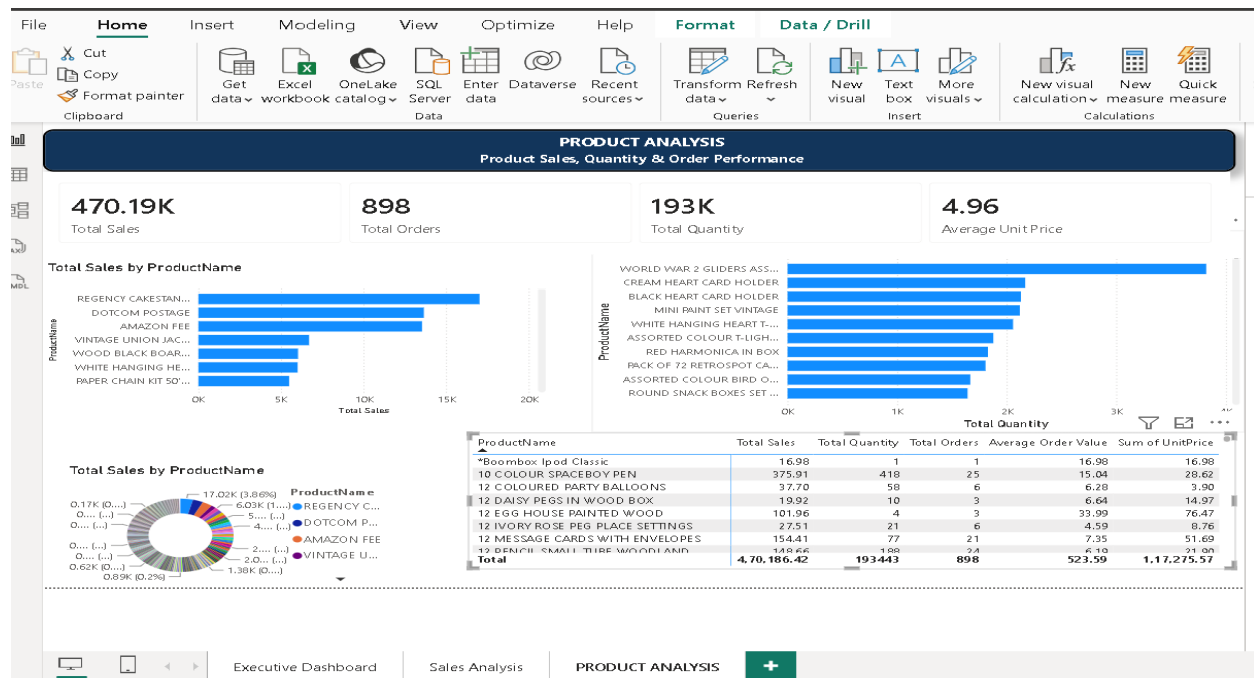
22. PAGE 3 — PRODUCT ANALYSIS

Title

PRODUCT ANALYSIS

Subtitle

Product Sales, Quantity & Order Performance



23 CONCLUSION

The Retail Business Analytics and Sales Performance Dashboard successfully demonstrates an end-to-end business intelligence solution.

The project begins with raw retail transaction data and processes it through MySQL for data cleaning, transformation, and structured storage. A star-schema model is developed to organize transactional and dimensional information efficiently.

Power BI is then used to connect to the database, create relationships, develop DAX measures, and build interactive dashboards. The solution provides executive, sales, product, customer, geographic, and forecasting analysis.

Additional capabilities such as What-If analysis and Row-Level Security improve the analytical and security features of the system. Publishing the solution to Power BI Service allows authorized users to access the report and supports centralized business reporting.

Overall, the project demonstrates how raw retail data can be transformed into meaningful business intelligence that supports data-driven decision-making, sales optimization, product management, customer analysis, and strategic planning.

Learning Outcomes

After completing the Retail Business Analytics and Sales Performance Dashboard project, the following learning outcomes were achieved:

1. Learned how to import and manage large retail datasets using MySQL Workbench.
2. Gained practical knowledge of SQL data cleaning, transformation, filtering, and validation.
3. Learned how to design a star schema using fact and dimension tables.
4. Developed skills in connecting MySQL with Power BI and preparing data using Power Query.
5. Learned to create DAX measures for KPIs such as Total Sales, Total Orders, Total Customers, Average Order Value, and Sales Growth.
6. Gained experience in designing professional and interactive Power BI dashboards using charts, cards, slicers, and filters.
7. Learned how to perform sales, product, customer, and geographical analysis using business intelligence techniques.
8. Gained practical understanding of sales forecasting and What-If analysis for business decision-making.
9. Learned how to implement Row-Level Security (RLS) to control access to business data.
10. Developed knowledge of publishing and sharing reports using Power BI Service and maintaining project files using GitHub.

Overall, the project provided practical experience in the complete end-to-end Business Intelligence workflow, from raw data collection and database management to data visualization, analysis, and business decision-making.

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Github:https://github.com/Balajivallepu/Data_Analytics_for_Bussiness/tree/main/Final_project

GITHUB_PPPIX:https://github.com/Balajivallepu/Data_Analytics_for_Bussiness/blob/main/Final_project/final_project_updated.pbix